



Lakshya JEE 2.0 (2024)

Application of Derivative

DPP-08

- The value of 'C' in Rolle's theorem for the function $f(x) = \cos \frac{x}{2}$ on $[\pi, 3\pi]$
 - 0
 - 2π
 - $\frac{\pi}{2}$
 - $\frac{3\pi}{2}$
- Rolle's theorem is applicable in the interval $[-2, 2]$ for the function
 - $f(x) = x^3$
 - $f(x) = 4x^4$
 - $f(x) = 2x^3 + 3$
 - $f(x) = x - |x|$
- If Rolle's theorem holds for the function $f(x) = x^3 + bx^2 + cx$, $1 \leq x \leq 2$ at the point $x = \frac{4}{3}$ then:
 - $b = 8, c = -5$
 - $b = -5, c = 8$
 - $a = -\frac{2}{3}, b = 1$
 - $a = \frac{2}{3}, b = 1$
- The value of 'C' in Lagrange's mean value theorem for $f(x) = \log x$ on $[1, e]$ is:
 - $\frac{e}{2}$
 - $e - 1$
 - e^{-2}
 - $1 - e$
- The value of 'C' in Lagrange's mean value theorem for $f(x) = x(x-2)^2$ in $[0, 2]$ is
 - 0
 - 2
 - $\frac{2}{3}$
 - $\frac{3}{2}$
- Let f be differentiable for all x, if $f(1) = -2$ and $f'(x) \geq 2$ for $x \in [1, 6]$, then:
 - $f(6) \geq 8$
 - $f(6) < 8$
 - $f(6) < 5$
 - $f(6) = 5$
- The function $f(x) = x^3 - 6x^2 + ax + b$ satisfy the conditions of Rolle's Theorem in $[1, 3]$ the value of 'a' and 'b' are:
 - $a = 11, b \in \mathbb{R}$
 - $a = -6, b = 11$
 - $a \in \mathbb{R}, b = 11$
 - $a = 6, b = -11$
- If $2a + 3b + 6c = 0$, ($a, b, c, \in \mathbb{R}$) then the Quadratic equation $ax^2 + bx + c = 0$ has:
 - At least one root in $[0, 1]$
 - At least one root in $[2, 3]$
 - At least one root in $[4, 5]$
 - None of these
- If f and g are differentiable function & in $[0, 1]$. Satisfying $f(0) = 2 = g(1), g(0) = 0$ and $f(1) = 6$, then for some $C \in]0, 1[$
 - $f'(C) = g'(C)$
 - $f'(C) = 2g'(C)$
 - $2f'(C) = g'(C)$
 - $2f'(C) = 3g'(C)$
- The equation $2 \tan x + 5x - 2 = 0$ has
 - no solution in $[0, \pi/4]$
 - at least one real solution in $[0, \pi/4]$
 - two real solution in $[0, \pi/4]$
 - none of these
- Let $f(x) = 5x^8 + ax^5 + bx^2 - cx - 10$ where $a, b, c \in \mathbb{R}$ and $f(2) = 5, f(4) = -5$, then minimum number of real roots when the equation $f(x) = 0$ can have is
 - 2
 - 3
 - 4
 - 5



Note: Kindly find the Video Solution of DPPs Questions in the DPPs Section.

Answer Key

1. (2)
2. (2)
3. (2)
4. (2)
5. (3)
6. (1)

7. (1)
8. (1)
9. (2)
10. (2)
11. (3)



PW Web/App - <https://smart.link/7wwosivoicgd4>

Library- <https://smart.link/sdfez8ejd80if>